

SKILL 7

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#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/types.h>
#include <limits.h>

#define MAX_INPUT 1024

int find_command(const char *command, char *result)
{
    char *path = getenv("PATH");

    if (path == NULL)
    {
        return 0;
    }

    char *path_copy = strdup(path);

    if (path_copy == NULL)
    {
        return 0;
    }

    char *dir = strtok(path_copy, ":");

    while (dir != NULL)
    {
        char full_path[PATH_MAX];

        snprintf(full_path, sizeof(full_path),
                 "%s/%s", dir, command);

        if (access(full_path, X_OK) == 0)
        {
            strcpy(result, full_path);
            free(path_copy);
            return 1;
        }
    }
}
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        dir = strtok(NULL, ".");
    }

    free(path_copy);
    return 0;
}

int main()
{
    char command[MAX_INPUT];
    char executable_path[PATH_MAX];

    printf("Enter command: ");
    fflush(stdout);

    if (fgets(command, sizeof(command), stdin) == NULL)
    {
        return 1;
    }

    command[strcspn(command, "\n")] = '\0';

    if (strlen(command) == 0)
    {
        printf("No command entered.\n");
        return 1;
    }

    printf("\nSearching PATH for: %s\n", command);

    if (!find_command(command, executable_path))
    {
        printf("Command not found: %s\n", command);
        return 1;
    }

    printf("Executable found: %s\n", executable_path);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

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}

if (pid == 0)
{
    printf("Child Process:\n");
    printf("PID: %d\n", getpid());
    printf("Executing: %s\n", executable_path);

    execl(executable_path, command, (char *)NULL);

    perror("exec");
    exit(1);
}
else
{
    int status;

    printf("Parent Process:\n");
    printf("PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    if (waitpid(pid, &status, 0) == -1)
    {
        perror("waitpid");
        return 1;
    }

    printf("\nChild process finished.\n");

    if (WIFEXITED(status))
    {
        printf("Child exited normally.\n");
        printf("Exit status: %d\n",
            WEXITSTATUS(status));
    }
    else if (WIFSIGNALED(status))
    {
        printf("Child terminated by signal: %d\n",
            WTERMSIG(status));
    }
}

return 0;
}

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adithya@LAPTOP-07A78KDI: ~$ nano process_path.c
adithya@LAPTOP-07A78KDI: ~$ gcc process_path.c -o process_path
adithya@LAPTOP-07A78KDI: ~$ ./process_path
Enter command: LS

Searching PATH for: LS
Command not found: LS
adithya@LAPTOP-07A78KDI: ~$ ./process_path
Enter command: pwd

Searching PATH for: pwd
Executable found: /usr/bin/pwd

Parent Process:
PID: 893
Child PID: 894

Child Process:
PID: 894
Executing: /usr/bin/pwd
/home/adithya/OS

Child process finished.
Child exited normally.
Exit status: 0
adithya@LAPTOP-07A78KDI: ~$ ./process_path
Enter command: hello123

Searching PATH for: hello123
Command not found: hello123
adithya@LAPTOP-07A78KDI: ~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.8.0_x64__8wekyb3d8bbwe:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath
:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/W
INDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/neofetch-win:/mnt
/c/Program Files/Git/cmd:/mnt/c/Users/Adithya/AppData/Local/Programs/Python/Python311:/mnt/c/Users/Adithya/AppData/Local/Programs/Python/Python311/Scripts/
:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/Docker/Docker/resources/bin:/mnt/c/Users/Adithya/Downloads:/mnt/c/Progr
am Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Adithya/.local/bin:/mnt/c/Users/Adithya/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/Adithya/AppDat
a/Local/Microsoft/WindowsApps:/mnt/c/Users/Adithya/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Adithya/AppData/Local/Python/bin:/mnt/c/Users/A
dithya/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Program Files (x86)/MongoDB Atlas CLI:/snap/bin
adithya@LAPTOP-07A78KDI: ~$ waitpid(pid, &status, 0);
-bash: syntax error near unexpected token `pid,'
adithya@LAPTOP-07A78KDI: ~$ waitpid(pid, &status, 0);

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adithya@LAPTOP-07A78KDI: ~$ waitpid(pid, &status, 0);
-bash: syntax error near unexpected token `pid,'
adithya@LAPTOP-07A78KDI: ~$ if (WIFEXITED(status))
-bash: syntax error near unexpected token `status'
adithya@LAPTOP-07A78KDI: ~$ WEXITSTATUS(status)
-bash: syntax error near unexpected token `status'
adithya@LAPTOP-07A78KDI: ~$ ./process_path
Enter command: WEXITSTATUS(status)

Searching PATH for: WEXITSTATUS(status)
Command not found: WEXITSTATUS(status)
adithya@LAPTOP-07A78KDI: ~$ wsl
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
adithya@LAPTOP-07A78KDI: ~$ cd os_experiment
-bash: cd: os_experiment: No such file or directory
adithya@LAPTOP-07A78KDI: ~$ |

```